TrialGuard AI
Clinical Trial Risk Monitor & Protocol Deviation Detector

PROJECT & SUBMISSION IDENTIFICATION

Team Name: TrialGuard Squad
Team Members: Daksh Soni (Lead), Visham, Het, Manav
Event: IBM Bob AI Innovation Hackathon 2026 (AI Track)
Application Type: Standalone Full-Screen Web Application (React + FastAPI)
Product Tagline: "Detect Earlier. Prioritize Smarter. Act Faster."
Positioning: Enterprise Clinical Decision-Support Prototype using Synthetic Trial Telemetry

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The Clinical Trial Monitoring Challenge
Manual site monitoring creates operational blind spots and delayed risk discovery

MONITORING BOTTLENECK FLOW

Patient Records + Visit Telemetry + Site Data
| (data ingestion)
Large Volume of Multi-Site Clinical Data
| (manual review bottleneck)
Hard-to-Detect Protocol Deviations (Late Visits, Dosing Errors, Prohibited Meds) V

Delayed Risk Identification & Reactive Site Audits

High Telemetry Volume

Hundreds of patient visits across multiple sites create massive
manual inspection fatigue.

Delayed Audit Signals

Protocol deviations remain hidden until periodic site audits, risking subject safety.

Unprioritized Risk

Reviewing records sequentially lacks explainable site-level risk scoring for clinical monitors.

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Introducing TrialGuard AI
An AI-assisted clinical trial monitoring web application for real-time risk intelligence

TrialGuard AI is a standalone full-screen web application designed for clinical trial monitors and quality managers. It evaluates clinical trial records against protocol rules, detects deviations, classifies severity, calculates an explainable 0-100 site risk score, and assists in generating standardized CAPA reports.

END-TO-END CLINICAL INTELLIGENCE WORKFLOW
Clinical Protocol -> Patient Data -> Protocol Comparison -> Deviation Detection -> Severity Rating -> Site Risk Score -> AI Copilot Explanation -> CAPA Report

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What TrialGuard AI Does
Six core capabilities built for proactive clinical trial risk oversight

1. Clinical Trial Dashboard
Macro overview tracking total patients, sites, active deviations, and
top critical risk sites.

2. Protocol Deviation Detector
Automated evaluation of missed visits, late visits, incorrect dosing, prohibited meds, & missing labs.

3. Severity Classification
Dynamic categorization of deviations into Major, Minor, and
Administrative severity tiers.

4. Explainable Site Risk Score
Weighted 0-100 risk scoring with transparent factor attribution (Low, Medium, High, Critical).

5. IBM Bob AI Copilot
Database-grounded conversational assistant for telemetry lookup, risk queries, & CAPA assistance.

6. CAPA Report Generator
Automated generation of issue summaries, root cause analyses,
corrective actions, & PDF exports.

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Technical Architecture

Full-stack decoupled Web Application stack with grounded AI telemetry

FULL-STACK WEB APPLICATION ARCHITECTURE

[Presentation Layer] TrialGuard AI Web Application (React 18 + TypeScript + Tailwind CSS + Recharts)
| REST API (JSON)
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[Backend REST Service Layer] FastAPI Framework (Python + Pydantic Schemas)
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SQLite Database Rule Engine AI Copilot Layer
(SQLAlchemy ORM) (Deviation Detector) (watsonx.ai REST + Deterministic Fallback)
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[Intelligence Output] CAPA Report Generator -> PDF Export / DB Record

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From Trial Data to Actionable Intelligence
Horizontal process pipeline executing automated protocol evaluation

STEP 1: Ingestion
Synthetic trial records stored in SQLite database.

STEP 2: Evaluation
Visit telemetry checked against protocol rules.

STEP 3: Storage
Deviations persisted with patient, site, & visit
linkage.

STEP 4: Severity
Dynamic classification (Major, Minor, Admin).

STEP 5: Risk Score
Weighted formula computes 0-100 site risk score.

STEP 6: AI Copilot
Database-grounded Q&A; answers telemetry queries.

STEP 7: CAPA
Automated Root Cause & Preventive Action generation.

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Inside TrialGuard AI (Web Application Interface)
Full-screen web dashboard user journey with actual implemented frontend views

CLINICAL MONITOR USER JOURNEY

Full-Screen Web Application User Journey:
1. Dashboard: Macro trial oversight (Patients, Sites, Active Deviations, Top Critical Site Alert).
2. Sites Overview: Site risk cards & filterable list (Risk score, major deviation count, risk factor drivers).
3. Deviation Details: Detailed master register of all protocol violations with patient/visit context.
4. AI Copilot: Grounded queries (e.g. "Why is SITE-003 high risk?" -> Answers: 100.0/100 CRITICAL).
5. CAPA Generator: Automated Root Cause Analysis & Preventive Action report export.

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IBM Bob as Our AI Development Partner
AI-assisted software development & database-grounded AI Copilot architecture

1. DEVELOPMENT PARTNER ROLE

IBM Bob Development Lifecycle Support:
- Architecture & Structure: System decomposition into React, FastAPI, & SQLAlchemy ORM.
- Feature & API Implementation: Rule-based detector logic & weighted risk scoring algorithms.
- Debugging & Verification: Automated full-stack verification test scripts.
- Documentation & Audit: Submission manifest validation and final repository compliance audit.

2. GROUNDED COPILOT ARCHITECTURE

AI Copilot Grounding Architecture:
User Question -> TrialGuard AI Copilot -> Current Trial SQLite DB -> Grounded Context -> watsonx.ai REST Integration (when configured) / Deterministic Fallback (when credentials unconfigured) ->
Zero-Hallucination AI Response

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Validated Working Prototype & Benchmark
Empirical verification results from local automated test suite

EMPIRICAL VALIDATION RESULTS

Verified Project Metrics & System Results:
- 5 Clinical Trial Sites Monitored: SITE-001 through SITE-005 successfully seeded.
- 50 Enrolled Patients: 10 subjects per site across 250+ scheduled visits.
- Dynamic Risk Calculation: SITE-003 correctly computed at 100.0 / 100 - CRITICAL.
- Protocol Deviation Detector: 100% detection accuracy across all 5 protocol rule types.
- Database-Grounded AI Copilot: 100% telemetry verification pass rate.
- CAPA Report Generator: Automated Root Cause Analysis & PDF exporter verified.
- Frontend Build Status: PASS (`npm run build` compiled cleanly).
- Backend API Tests: PASS (FastAPI endpoints & ORM queries verified).
- Repository Validation Workflow: PASS (Passed all submission criteria).

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From Detection to Action

Transforming clinical trial risk oversight with actionable AI intelligence

CLINICAL IMPACT

Clinical Impact:
- Earlier Risk Identification: Flags high-risk sites before safety audits fail.
- Faster Deviation Review: Automated rule evaluation reduces monitor review cycles.
- Explainable Prioritization: Transparent mathematical factor weighting.
- Structured CAPA Support: Instant standardized root-cause documentation.

FUTURE ROADMAP

Future Vision & Roadmap:
- EDC Integration: Medidata Rave & Veeva Vault connectors.
- Advanced Analytics: Unstructured clinical note NLP inspection.
- Enterprise Security: HIPAA-compliant cloud & role-based access.
- Regulatory Workflows: Validated audit logging for GxP compliance.

Disclaimer: TrialGuard AI is a prototype decision-support demonstration and is not validated clinical, medical, patient-safety, or regulatory software.

TrialGuard AI -- "Detect Earlier. Prioritize Smarter. Act Faster."

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